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The attitude–behaviour gap in sustainable tourism

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ABSTRACT

This study investigates why people who actively engage in environmental protection at home engage in vacation behaviour which has negative environmental consequences, albeit unintentionally. The environmental activists participating in the study were highly aware of the negative environmental consequences of tourism in general, but all displayed an attitude–behaviour gap which made them feel uncomfortable. Participants did not report changing their behaviour; instead, they offered a wide range of explanations justifying their tourist activities. Gaining insight into these explanations contributes to our understanding of why it is so difficult to motivate people to minimize the negative environmental impacts of their vacations, and represents a promising starting point for new interventions to reduce environmentally unsustainable tourism behaviours.

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Introduction

Tourists generally have positive attitudes towards the environment, and do not wish to behave in ways that negatively impact the environment (Dolnicar, 2004; Wurzinger & Johansson, 2006). Social psychological theories explaining why humans behave in certain ways, such as the theory of planned behaviour (Ajzen, 1985), postulate that attitudes—among other factors—affect behaviour. Yet, in the context of environmentally sustainable behaviour, having a positive attitude does not emerge as a good predictor of making environmentally sustainable vacation choices. A distinct attitude–behaviour gap exists in environmentally sustainable tourism (Becken, 2004; Bergin-Seers & Mair, 2009).

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As part of investigating different research questions in a range of study contexts, several studies have identified possible reasons for this gap, including: claiming that there are no alternatives to current behaviours; that other issues are of greater importance (Becken, 2007; Buckley, 2011; Lorenzoni, Nicholson-Cole, & Withmarsh, 2007); using escape and relaxation as an excuse for disregarding environmental considerations (Wearing, Cynn, Ponting, & McDonald, 2002); not having the information required to choose vacation options that come at a low environmental cost (Juvan & Dolnicar, 2013); buying offsets or using credits of smaller footprints from everyday life, or behaving in an environmentally friendly way at home (Becken, 2007; Buckley, 2011); being too busy to change one's behaviour (Lorenzoni et al., 2007); blaming others (Lorenzoni et al., 2007; Miller, Rathouse, Scarles, Holmes, & Tribe, 2010); having faith in technological solutions (Gössling, Haglund, Kallgren, Revahl, & Hultman, 2009; Lorenzoni et al., 2007); denying responsibility (Gössling et al., 2009); displacing responsibility (Lorenzoni et al., 2007; Miller et al., 2010); emphasizing the job creation benefits of carbon emitting industries (Becken, 2007); and arguing that there is negligible impact from personal behaviour (Gössling et al., 2009; Lorenzoni et al., 2007). None of these studies, however, identifies a comprehensive list of reasons for the attitude–behaviour gap in environmentally sustainable tourism, or develops a systematics of reasons. This is the aim of the present study.

Specifically, the present study aims to produce a better understanding of the attitude–behaviour gap in the context of environmentally sustainable tourism. The investigation focuses on environmental activists because they are known to have pro-environmental attitudes (Stern, Dietz, Abel, Guagnano, & Kalof, 1999) and have a demonstrated willingness to engage in behaviours that help the environment. We may expect, therefore, that environmental activists will demonstrate the smallest attitude–behaviour gap, if any at all. This paper investigates the following research questions:

(1) Are environmental activists aware of the negative environmental consequences of their vacation behaviour?
(2) Is there an attitude–behaviour gap among environmental activists in the tourism context?
(3) If so, how do environmental activists feel about this attitude–behaviour gap? And do they attempt to reduce the gap?

The study primarily contributes to the understanding of environmentally unsustainable vacation behaviour. The insights gained may form the basis of developing targeted interventions in the future which aim at reducing vacation behaviours that come at high environmental cost.

Theoretical background

Several theories and concepts have been used in the past to explain behaviours that cause harm to the environment and therefore need to be considered as possible bases for the present study, which focuses on the attitude–behaviour gap in the context of environmentally sustainable tourism.

Theory of planned behaviour

The theory of planned behaviour (Ajzen, 1985), an extension of the theory of reasoned action (Fishbein & Ajzen, 1975), postulates that attitudes, social norms and perceived behavioural control affect people's intentions to behave in certain ways which, in turn, lead to actual behaviour. The theory of planned behaviour has frequently been used as a basis for investigating environmentally sustainable behaviour in general (Anable, Lane, & Kelay, 2006; Bamberg, Ajzen, & Schmidt, 2003; Chen & Tung, 2010; Fielding, McDonald, & Louis, 2008; Kalafatis, Pollard, East, & Tsogas, 1999; Shaw, Shiu, & Clarke, 2000) and environmentally sustainable tourism behaviour in particular (Han, Hsu, & Sheu, 2010; Han & Kim, 2010; Ong & Musa, 2011), showing great promise in explaining behavioural intentions. Critics of the theory of planned behaviour argue, however, that behavioural intentions do not translate into behaviour, and several empirical studies have demonstrated that this link is indeed relatively weak (for example, Bergin-Seers & Mair, 2009; Bickmann, 1972; McDonald, Oates, Alevizou, Young, & Hwang, 2012; McKercher & Tse, 2012).

Specific criticism has also emerged from studies investigating environmentally sustainable behaviours. For example, Prillwitz and Barr (2011) found that "green attitudes (towards the environment and sustainability in general) do not influence tourist travel significantly" (p. 1595). Bickmann (1972) shows experimentally that 94% of respondents believe that "picking up litter is everyone's responsibility", but only 1.4% actually picked up litter when exposed to it, and concludes that "environmental problems will not be solved by simply influencing verbally expressed attitudes" (p. 324). Anable et al. (2006) conclude that the theory of planned behaviour is "too simplistic for a study of travel behaviour and climate change" (p. 61) because it has typically been used to describe behaviours with two simple behavioural alternatives.

The theory of planned behaviour has contributed significantly to understanding drivers of human behaviour. It is valuable to the present research in that it points to social norms and perceived behavioural control as possible reasons for attitudes not translating into behaviour.

Attribution theory

Attribution theory (Heider, 1985) postulates that people have two fundamentally different ways of explaining the causes of their own behaviour as well as events affecting them. When using internal or personal attribution, people see themselves as the cause. When using external situational attribution, people see reasons not related to them as the cause. Attribution theory lends itself as a basis for studies of environmentally sustainable tourism because people can either attribute the cause of climate change and other direct or indirect negative environmental consequences of tourism to themselves or to others. Attributing these issues to others means not having the power to change anything, and thus implies that one cannot be held accountable for any negative environmental impacts.

Attribution theory has been explicitly used in tourism research to understand to whom tourists attribute vacation experiences (Jackson, White, & Schmierer, 1996), and to investigate how hotel guests' willingness to pay for the environmentally sustainable credentials of hotels is affected by their attribution of why hotels communicate those credentials (Kang, Steinb, Yoojoung, & Lee, 2012). It has hitherto not been used to understand environmentally unsustainable vacation behaviour.

In terms of the potential to form a theoretical basis for investigating the attitude–behaviour gap, attribution theory has its merits, because it offers one possible explanation for such a gap: the fact that people simply do not see themselves to be the cause of the problem, and therefore do not see behavioural change on their part to be part of the solution.

The value-belief-norm theory of environmentalism

Stern's (2000) value-belief-norm theory of environmentalism postulates relationships between a person's values and beliefs about the environment, one's responsibility for environmental conditions and personal norms relating to the environment, which, in turn, cause pro-environmental behaviours.

Beliefs are understood to be formed and modified throughout a person's life (Rokeach, 1968), and thus represent a potential target for interventions that aim to induce behavioural change. The value-belief-norm theory of environmentalism postulates that two specific beliefs affect pro-environmental behaviours via norms: 1) the awareness of consequences of behaviour on the things people value; and 2) the ascription of responsibility, indicating that people believe they are responsible for protecting that which they value so highly. Interventions aimed at those two beliefs have been successful in increasing the level of pro-environmental behaviour (Hopper & Nielsen, 1991; Stern et al., 1999; Van Liere & Dunlap, 1978).

The value-belief-norm theory has not been used to date as the basis for the study of environmentally sustainable tourism, despite the fact that both of the key constructs of the theory—awareness of consequences and ascription of responsibility—have been individually studied in the context of the environmental burden of tourism activity (Becken, 2004, 2007; Gössling et al., 2009; McKercher & Prideaux, 2011; Miller et al., 2010), concluding overwhelmingly that people ascribed little responsibility to themselves in terms of their contribution to climate change due to tourism activity.

We may conclude that value-belief-norm theory is valuable because it points to two specific reasons that may explain the attitude–behaviour gap under study: the fact that people may think that

taking a vacation does not have a negative impact on the environment, and the fact that they may not feel responsible for the problem, and thus similarly not responsible for contributing to the solution.

Cognitive dissonance theory

Cognitive dissonance theory (Festinger, 1957) postulates that people experience psychological discomfort when there is an inconsistency between "cognitions (attitudes, beliefs, values, opinions, knowledge) about themselves, about their behaviour and about their surroundings" (p. 9). Cognitive dissonance can manifest in many ways. Festinger (1957) describes it as frustration or disequilibrium, Davidson and Keisler (1964) suggest the feeling of dissonance makes people equivocal, confused, unclear or oblique, and, according to Sweeney, Hausknecht, and Soutar (2000), people can react by feeling anxious, uncertain or doubtful.

For cognitive dissonance to occur, a person must have the desire to achieve a certain outcome and must value that outcome. Festinger (1957) suggests that it is necessary to distinguish different degrees or magnitudes of dissonance. Similarly, Soutar and Sweeney (2003) state that "cognitive dissonance is not a specific condition but it rather exists from lesser to greater extent, at various stages in decision-making" (p. 231). The greater the dissonance, the greater the intensity of the action to reduce the dissonance, and the greater the avoidance of situations that increase dissonance.

People respond to cognitive dissonance by adjusting either beliefs or behaviours, such that "states of dissonance are transformed into states of consonance and the inconsistencies are eliminated" (Kassarjian & Cohen, 1965, p. 56). This can be achieved either by changing behaviour to align with beliefs or by changing beliefs to align with behaviour. More recently Stone and Fernandez (2008) found evidence that people will reduce the tension by changing behaviours if they are given the opportunity to publicly advocate and commit themselves to correct behaviours and are privately reminded of their own failures to perform the target behaviour in the recent past; if reminders are public, people are more likely to change beliefs to cope with the tension.

Cognitive dissonance has been used in the general context of environmentally sustainable behaviour. For example, Thorgesen (2004) concludes that "the interaction between moral norms and similarity perceptions in determining the level of behavioural consistency strongly supports the proposition that the desire to avoid cognitive dissonance is driving this process" (p. 101).

In the context of environmentally sustainable tourism behaviour, no study has yet used cognitive dissonance as the basis of its investigation, but a few researchers report observing cognitive dissonance or hypothesize that cognitive dissonance may be occurring; for example, Hares, Dickinson, and Wilkes (2010) found that people do not modify their vacation behaviour in view of climate change, and hypothesise that "it is possible they may have aligned their attitudes towards holidays and climate change to be consistent with their behaviour" (p. 472). Similarly, Miller et al. (2010) found that people who are concerned about the environment but still travel experience emotional dissonance and report feeling bad. Miller et al. (2010) note that tourists may be avoiding or reducing dissonance by rejecting information about the negative environmental impact of their tourism-related activities.

Cognitive dissonance theory focuses specifically on explaining the attitude–behaviour gap, which stands at the centre of the present study, making it the most suitable theory to form the basis of the investigation. Additional support for the choice of cognitive dissonance theory is provided by several studies, which demonstrate that cognitive dissonance can be used to change environmentally sustainable behaviours: Dickerson, Thibodeau, Aronson, and Miller (1992) show that participants who were reminded of their past wasteful water use and made a commitment to reduce water use took significantly shorter showers than subjects who were merely reminded of past wasteful behaviour or only made the public commitment. The authors conclude that "the findings have implications for using cognitive dissonance as means of changing behavior in applied settings, especially those in which people already support the desired goal, but their behavior is not consistent with those beliefs" (p. 841). Similarly, Kantola, Syme, and Campbell (1984) reduced household energy consumption by informing respondents about the discrepancy between their stated attitudes towards energy use and their high energy use. Aitken, McMahon, Wearing, and Finlayson (1994) found "that statistically significant reductions in residential water consumption can be achieved by the application of

simple cognitive dissonance and feedback information, at least in the short term. Households receiving both dissonance and feedback treatments exhibited the greatest reductions" (p. 154), suggesting that the "added effect of placing the subject in a dissonant situation is more effective in encouraging a reduction in water consumption" (p. 154). Stone, Aronson, Crain, Winslow, and Fried (1994) provide evidence that people who were made aware of the gap between what they say about safer sex and how they practice it are over two times more likely to follow through on their high intentions for safer sex and acquire condoms. Fointiat (2004) reported that inducing cognitive dissonance leads to changes in driving behaviour. Results of the experiment about asking people to volunteer to become safer drivers show that significantly more participants in the dissonance induction condition volunteered to have their driving monitored than in the control group.

Methodology

As Miller (2003) points out in the context of reviewing a past survey based research on sustainable consumerism in tourism, "a weakness of much of this research is the distinction between what survey respondents say and what they actually ask for or do" (p. 19). In order to avoid this problem as much as possible, a qualitative research approach that included both prompted and unprompted questions was chosen. Throughout the interview, primarily open-ended questions were used to ensure the unbiased identification of a wide range of beliefs that were potentially instrumental in the re-establishing cognitive consonance. Woodside and Lysonski (1989) argue that such unaided awareness response measures are associated strongly with behaviour. Open-ended questions also have the advantage that the interviewer is in control of the question sequence, allowing the slow revelation of the precise topic of investigation to the participants (Wagner, 2003).

Interviews were conducted with members of environmental organisations, because it was expected that their attitude–behaviour gap relating to environmentally sustainable vacation behaviour would be minimal (Karp, 1996; Randle & Dolnicar, 2006). Selecting participants based on actual behaviour, rather than their responses to a set of items on values or attitudes, is critical in order to avoid social desirability bias (Fisher, 1993), which would lead to people being incorrectly classified as highly pro-environmental.

Participants were recruited through organisations with a mission to protect the environment. Typical activities of members of those organisations include hands-on environmental work, such as planting trees and collecting litter, and activist tasks, such as organising and participating in pro-environment rallies, door knocking to collect signatures on petitions relating to environmental issues, fundraising, providing information about environmental issues on the Internet and recruiting additional members.

Data were collected in Australia and Slovenia because the researchers were physically located there. Participants were heterogeneous with respect to gender, age (ranging from 23 to 65), education level, occupation, country of origin, number of vacations undertaken per year and years of involvement in environmental activism. A detailed profile of all participants is provided in Table 1. To ensure that participants were members of the environmental organisations because they wanted to contribute to protection and conservation of the environment, they were asked why they chose to be actively involved in this organisation at the beginning of the interview. The responses to these questions—which are also provided in Table 1—confirm that they hold pro-environmental cognitions.

Neither the representatives of the organisations who were initially contacted, nor the participants themselves were informed about the *precise* purpose of the study, to avoid biasing responses. Instead, they were told that the study investigated how people plan their vacations and how they behave when on vacation.

The sample size was not set in advance; rather, the approach of seeking for the point of data saturation (Glaser & Strauss, 1967) was used. The point of data saturation is reached when additional interviews do not lead to additional insights. The data saturation approach is widely used in qualitative tourism research (for example, Guest, Bunce, & Johnson, 2006; Lumsdon & McGrath, 2011; Nimrod, 2008; and papers identified in the Appendix) . For the present study, this point was reached at 25 interviews. To double check that 25 respondents aligned with design choices of other qualitative researchers in tourism, a review of purely qualitative studies published in 2012 and 2013 in the *Annals*

Table 1
Participant profiles.

ID	Country of residence, country of origin	Age	Sex	Degree	Vacation trips per year	Job	Years of and reasons for being a member of an environmental organisation
Original sample							
WS1	Australia, Australia	27	F	Bachelors	6	Finance officer	Two years; passionate about environment; The Wilderness Society gives opportunity to actually do something; I want to protect environment in an active sense; stop pollution
WS2	Australia, Japan	30	F	Masters	4	Student	Three years; I want to get involved in environmental protection; interest in environmental issues; socialize with likeminded people
WS3	Australia, USA	23	F	Masters	3	Student	Four years; to get involved in helping people; nature; I did environmental science class and got interested
WS4	Australia, France	27	F	Masters	5	Unemployed	Three years; passionate about environment; want to make a difference
WS5	Australia, Australia	26	M	Masters	4	Public officer	Three years; share values of The Wilderness Society people and want to help them make a difference
WS6	Australia, Australia	53	F	Bachelors	5	Manager	10 years; passion for environment; wanted to make contribution; make sure environment is not destroyed
WS7	Australia, Australia	51	M	Bachelors	4	Public officer	15 years; love for nature; care for nature; discovery that wildlife is being destroyed
WS8	Australia, Australia	early 50s	F	Bachelors	3	Creative director	Five years; I wanted to voice my opinion: agreement with many ravens of The Wilderness Society; interest in nature and wildlife
WS9	Australia, Australia	23	F	Undergraduate diploma	20	Student	Eight years; angry about climate change; want to make a difference; need to protect environment and nature
WS10	Australia, Australia	29	F	Masters	1	Environmental management	10 years; love nature; love environment; we need to save the nature and environment; preserve it for future generations
WS11	Australia, France	31	F	Masters	2	Marketing manager	Five years; love nature; respect nature and its inhabitants; concerned about environmental issues;

(*continued on next page*)

Table 1 (*continued*)

ID	Country of residence, country of origin	Age	Sex	Degree	Vacation trips per year	Job	Years of and reasons for being a member of an environmental organisation
							like to do something about it
HU1	Slovenia, Slovenia	27	F	Bachelors	3	Event organizer, journalist	Two years; love to promote environmental issues; love to educate others; respect nature and wildlife; recognize environmental issues
HU2	Slovenia, Slovenia	35	F	Bachelors	1 or 2	Translator	Two years; interested in environmental problems; need to give back to the society and nature; love nature and pristine areas
BC2	Australia, Australia	43	F	Post-doctoral degree	2	Academic	12 years; feel morally obliged to protect environment; love pristine areas
BC1	Australia, Australia	52	F	Undergraduate diploma	6	Civil servant	17 years; giving back to community; a good way to protect what is beautiful; love vegetation
BC3	Australia, Australia	26	F	Bachelors	1 or 2	Unemployed	1.5 years; bush regeneration is interesting, relaxing; passion for nature and biodiversity which is being threatened; need to help
BC4	Australia, Australia	24	F	Bachelors	1	Student	Two years; personal and professional interest in nature; love to help with conservation; get to meet people with same interest; participate in protection of plants
BC5	Australia, Australia	26	F	Bachelors	2	Environmental engineer	Two years; to learn about nature; preserve nature; protect environment; to learn about environmental system
BC6	Australia, Australia	65	F	Bachelors	2	Retired	30 years; love the bush; love wildlife; hate how it's being destroyed; enjoy being in the bush
OT	Slovenia, Slovenia	33	F	Masters	2	Singer, event organizer	Three years; to promote responsibility towards society, nature, animals; upset about the way people treat nature; aware of the environmental issues and the need to do something about it
LG	Australia, Thailand	28	F	Masters	3	Student	10 years; it's outrageous how things are developing about waste pollution, water pollution; destruction of nature; love to be in nature; understand that we need to protect environment

Table 1 (*continued*)

ID	Country of residence, country of origin	Age	Sex	Degree	Vacation trips per year	Job	Years of and reasons for being a member of an environmental organisation
UQCG	Australia, Australia	36	F	Post-doctoral degree	2	Researcher	21 years; to raise awareness about environmental issues; to help redirect society from the destructive path; appreciation of nature and wildlife
UQEC	Australia, Australia	29	F	Bachelors	3	Researcher	Five years; passionate about environmental issues; concerned about the state of the environment; worry and guilt which I carry with me; big part of who I am
AYCC	Australia, Australia	31	F	Bachelors	8	Circus performer	14 years; it's a meaningful way to help environment and nature, society; passionate about the values of environmental organisations
AYCC	Australia, Australia	28	M	Undergraduate diploma	1	IT developer	Two years; passionate about doing something about climate change; appreciate nature; want to help protect nature for future generations; aware that climate change is escalating and threatens our society
Validation sample							
VAL1	Canada, Australia	35	F	Masters	12	Research administrator	5 years; climate change is an issue; we need to be more aware; respect for the environment, responsibility to share knowledge
VAL2	Australia, Australia	48	F	Undergraduate degree	4	Project administrator	18 years; appreciate nature; have a sense of connection with country and our planet; love nature; concerned about environment; worried about pollution, morally obliged to do something
VAL3	Australia, Australia	26	F	Bachelors	4	Water engineer	6 years; value earth and people on it; want to help solving water sustainability problems; share values of Greenpeace
VAL4	Australia, Australia	20	M	Undergraduate	2	Student	3 years; climate change is quite fundamental threat, it's the greatest challenge which affects my generation and future generations, and I have responsibility for my kids and future generations to do something about it
VAL5	Venezuela, Australia	39	F	PhD	6	Housewife	10 years; concerned about environment and climate change; appreciate nature; love wildlife
VAL6	Australia,	46	M	Bachelors	2	Project	20 years; aware of

(*continued on next page*)

Table 1 (*continued*)

ID	Country of residence, country of origin	Age	Sex	Degree	Vacation trips per year	Job	Years of and reasons for being a member of an environmental organisation
	Australia					coordinator	environmental issues; want to support organisations which are effective; something needs to be done
VAL7	Australia, Australia	31	F	Masters	2	Student	10 years; concerned about the future if we continue to damage the environment, nature and wildlife; feel responsible to promote change
VAL8	Australia, Australia	21	F	Undergraduate	2	Student	4 years; very concerned about climate change and how it will affect our future; I see that there is a solution; I want to help to get there
VAL9	Australia, Australia	34	F	Bachelors	2	Student	10 years; epic around climate and environment; concerned about where are we going in terms of ecology
VAL10	Slovenia, Slovenia	25	F	Bachelors	3	Student	3 years; disapprove environmentally harmful behaviour; have knowledge and feel obliged to share it; share Greenpeace values

of Tourism Research was conducted. Sample sizes for each study are provided in Table 2 in the Appendix. The average sample size of those studies is 28, with a minimum of five and a maximum of 65 respondents.

Interview locations were chosen by participants (for example, bar, park, office). Interviews lasted between 45 and 100 minutes. A semi-structured interview guide was used: respondents were first asked to talk about what was important to them when they went on vacation and which criteria they used when choosing a destination, accommodation, activities and tourism providers more generally. To make the questions less abstract, similar questions were asked about a specific recent vacation. A subset of respondents was asked directly whether they sometimes felt the tension between enjoying a vacation and knowing that it had negative environmental consequences, how it made them feel and how they dealt with this tension. The reason for asking only a subset of respondents direct questions is that they may have influenced the answers given by the study participants. It was important, therefore, to conduct interviews that did not aim directly at the constructs under study, to determine whether respondents not directly prompted would also report mechanisms re-establishing cognitive consonance. It is clearly indicated in the results section which comments were made by respondents who were directly asked about cognitive dissonance and which comments were made by respondents who were not prompted.

With the permission of participants, interviews were audio recorded and transcribed verbatim within a few hours of the interview. A contact summary report (Miles & Huberman, 1994) was written after each interview. The purpose of writing the contact summary reports was to record the specifics of the interview critical to data interpretation, such as nonverbal expressions, prolonged thinking, changing of ideas and interview flow. Summary reports and transcripts formed the basis of data analysis.

Data analysis was conducted in two stages: first, all statements reflecting explanations of the attitude–behaviour gap were identified and ordered into themes. Nineteen categories of explanations emerged. Next, similarities and dissimilarities between these explanations were used to group them into six higher-order categories (Corbin & Strauss, 2008). None of the possible explanations offered by

theories was included by default; rather, explanations were categorized based on their content only. This is in line with Glaser and Strauss's (1967, p. 37) recommendation that it is essential to first "ignore the literature of theory and fact on the area under study, in order to assure that the emergence of categories will not be contaminated by concepts more suited to different areas. Similarities and convergences with the literature can be established after the analytic core of categories has emerged". Only after completion of the data analysis process were resulting groups of beliefs compared with theoretically postulated reasons for attitude–behaviour gaps. Both levels of categories cover all empirical observations and allow independent assessment and agreement with proposed categorisation, which lends support to the credibility of the findings (Charmaz, 2006). To validate the findings, an additional ten interviews were conducted. Written transcripts of 82 explanations provided by these ten respondents were given to two raters. Raters independently categorized them into one of the six main and 19 sub-categories. The inter-rater agreement index (Kappa) was 0.881 ($p < .0.000$) at the level of 19 for sub-categories and 0.859 for the six main categories ($p < .0.000$). According to Landis and Koch (1977) Kappa values of between 0.81 and 1 indicate "almost perfect" agreement.

Results

Awareness of negative environmental consequences of vacation behaviour

Results indicate that, overall, the study participants were acutely aware of the negative consequences of climate change and environmental pollution, and felt very strongly about preventative action being required, as illustrated by the quotes below:

> *Horrifying, absolutely horrifying. If we don't make a difference to our polluting behaviours rapidly, immediately and globally, it is likely that human civilization will not survive in its current form* (prompted, female, 43 years old).

Study participants also demonstrated a general awareness of how tourism-related behaviour contributes to environmental pollution and climate change, especially awareness about the negative environmental impact caused by carbon emissions from transportation in general and aviation in particular:

> *Well, people fly too much. The impact of the airline industry has a tremendous impact on climate change and, of course, the more tourists are travelling, the more planes are in the sky* (unprompted, male 51 years old).

However, the negative consequences of tourism-related behaviours not directly related to transportation were also acknowledged by study participants, including the negative environmental impact of waste caused by tourism and damage done to pristine natural areas:

> *Waste pollution is particularly problematic in third world countries... tourists contribute... by going there* (unprompted, female, 33 years old).

> *In South America you can do tours of the tropical rainforest. When they constantly bring people through a certain path... the beauty of that path starts to decay and... it becomes trashed* (unprompted, female, 23 years old).

Interestingly, however, even among the group of participants in this study who were specifically selected as engaging in environmental action, there was some doubt and confusion about the precise negative effects that different tourism activities have on the environment:

> *We think that huge hotel development is less eco-friendly, but at the same time... it's concentrating a large number of... tourists... in one small area... so what is less or more eco-friendly... is a bit ambiguous* (unprompted, female, 27 years old).

> *I guess there is a whole thing with carbon emissions for the flight, but... people kind of offset flights now, so hmmm I'm not too sure about that* (unprompted, female, 27 years old).

Overall, based on the statements made by study participants, we may conclude that the environmental activists were aware of the environmental consequences of tourism. There may be some lack of clarity about the precise environmental impact of certain activities related to tourism, but overall, the participants were acutely aware of the potentially negative impacts of tourism on the environment.

The attitude–behaviour gap

Differences between study participants existed with respect to whether or not they openly admitted the attitude–behaviour gap or not. The following two quotes are representative of those study participants who were aware of the fact that their behaviour was not aligned with their pro-environmental attitudes:

I know this is not consistent with what I said before, but there is only that much I can do (male, 26 years old, unprompted).

I am aware that this is not how I actually feel and think about these things, but it was because of the kids – they love it (female, prompted, 36 years old).

It is evident from all interviews, however, that an attitude–behaviour gap existed in the context of vacation behaviour and its potential environmental impact. This finding is surprising, given that the sample was selected to consist of people who—in their everyday life—engaged in organised environmental protection or conservation action and who expressed pro-environmental cognitions in the interviews (see Table 1). Of all people, it would be such environmental activists who would be expected to have the smallest attitude–behaviour gap.

Cognitive dissonance, attempts at re-establishing consonance and beliefs used to do so

Only one respondent stated that they did not feel any tension about the attitude–behaviour gap. Many respondents—either unsolicited or in response to the question about perceiving a tension—reported experiencing what is likely to be cognitive dissonance. For example:

Something that uses a lot of resources and is wasteful makes me uncomfortable (prompted, female, 26 years old).

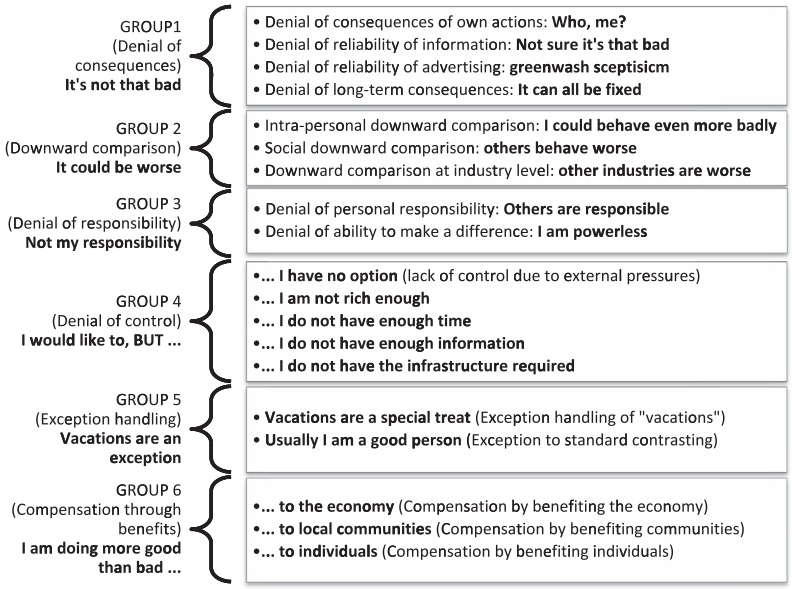


Fig. 1. Beliefs used to re-establish cognitive consonance.

It makes me feel upset and guilty and sometimes I don't enjoy things as much because of knowing that (prompted, female, 43 years old).

. . .disturbed me greatly. I didn't feel good about it (unprompted, female, 53 years old).

Participants offered a wide range of explanations for the attitude–behaviour gap. In total, six groups of beliefs emerged, which are summarized in Fig. 1 and described in detail below. As can be seen, some align directly with predictions made by the theory of planned behaviour, attribution theory and the value-belief-norm theory of environmentalism.

GROUP OF BELIEFS #1: It's not that bad (denial of consequences)

Beliefs in this group relate to awareness of consequences, one of the two key constructs postulated by the value-belief-norm theory of environmentalism to be a precursor to pro-environmental behaviour. These beliefs centre on the denial of the negative consequences on something valued, in this case, the environment. This denial can take a number of different forms, including denial of one's own actions or the denial of being part of the kinds of tourists who harm the environment. This subset of beliefs is therefore labelled **who, me?** and is illustrated in the following two statements:

I'm just not that kind of a tourist; it's not what I do. I simply explore natural places, walk around the city, try local foods and experience local things (unprompted, male, 26 years old).

We did not even do the mainstream stuff, like dirt bike riding. . . we try to do stuff that is not damaging. . . we don't do stuff like that (unprompted, female, 34 years old).

In many ways this represents a selective awareness of consequences characterised by the denial of negative consequences to the environment from those activities study participants engage in. Interestingly, these participants were not lacking in awareness of the negative environmental impacts of tourism in general, but they appeared to change their position when it came to their own behaviour. Given that they did not lack awareness, knowledge or understanding, this is indicative of belief adjustment for the purpose of re-establishing consonance.

A second belief that falls into this group is that of uncertainty about the negative consequences due to the questionable reliability of information provided to the general public, as illustrated in the following quotes, and labelled **not sure it's that bad**:

We never receive proper information on climate change, so I don't know what is true or untrue (unprompted, female, 27 years old).

Related to the uncertainly aspect is another belief which can be labelled **green wash scepticism**. It goes one step further from believing that there is insufficient evidence or information about the negative impacts of certain vacation-related activities, in that people actively question the motives of the providers of information related to the protection of the environment provided to them by the tourism industry through sales and advertising.

Environmentally certified tourism providers are not necessarily green; and not-certified are not necessarily harmful (prompted, female, 33 years old).

I fundamentally dispute offsetting, because it's a marketing gimmick (unprompted, male, 26 years old).

The final belief identified in the group of **its not that bad** beliefs was **it can all be fixed**, as illustrated in the following statements:

Earth has a recuperative power and, with the help of technology, it can recuperate quickly (prompted, female, 52 years old).

This statement exemplifies a specific belief that denies the consequences of environmental damage at a high level, by assuming that any environmental damage can be fixed, so even if some damage is caused, this is only a temporary problem. As a consequence of the belief in the fact that environmental

damage caused by tourism can be fixed, it follows that it is not such a big problem to behave in ways that have negative environmental consequences.

GROUP OF BELIEFS #2: It could be worse (downward comparison)

This second group of beliefs has in common with the first the mechanism of comparative judgment, specifically, downward comparison, as discussed by Festinger (1954) in the context of social comparison theory, a theory not originally considered as playing a role in environmentally sustainable tourism behaviour. Festinger postulates that, in an attempt to accurately define themselves, people compare themselves to others. Downward comparison is a specific mechanism that involves seeking out people who are in a worse situation, or, as in the case of pro-environmental behaviour, cause more damage to the environment, in order to feel better about themselves and their behaviour.

The responses provided in our study that relate to this mechanism indicate that a person's own behaviour, or tourism activity in general, is contrasted to other people's behaviour or the impact of other industries, in order to come to the conclusion that tourism is actually not that bad. The first subset of those beliefs results from an intra-individual comparison with other behaviours one could be engaging in that would be worse, and is therefore labelled **I could behave even more badly**:

I would travel all the time. I love travelling, but I don't travel that much because I believe that this would have a negative impact on the environment. And I mean that from the bottom of my heart. This is not a lie which would make me sound better! (prompted, female, 33 years old).

The fact that one could behave in a certain way, but does not, gives comfort in one's behaviour and helps to re-establish cognitive consonance. Similarly, the downward comparison with other people, either other tourists or locals, can be used to give this same kind of comfort (**others behave worse**):

I try to remind myself that we have done a lot more than an average Australian (prompted, female, 26 years old).

Locals... also use motorbikes – poor exhaustion systems – and that also causes tremendous impacts (unprompted, female, 35 years old).

Finally, the comparison can also occur at the industry, rather than the individual level, leading to the beliefs that **other industries are worse** and, as a consequence, supporting the tourism industry is comparatively harmless:

By traveling I harm, but so is the massive industrialisation and the mining business... their impact is greater than that of tourism (unprompted, female, 48 years old).

Tourism is just a puzzle in a big problem, so either we solve the whole puzzle or we won't do anything (prompted, female, 28 years old).

GROUP OF BELIEFS #3: It is not my responsibility (denial of responsibility)

This group of beliefs is tied closely to the construct of ascription of responsibility in the value-belief-norm theory of environmentalism, but encompasses both aspects of genuine **externalisation of responsibility** and aspects of deriving the idea of not being responsible from the feeling that one's behaviour cannot actually make a difference, and this state of **powerlessness** relieves one from any kind of responsibility. This also follows predictions from attribution theory when causes for events are externally attributed.

Yes, I do think how much fuel will this jet use to get me somewhere, but then again, this aircraft will take off with or without me anyway (unprompted, female, 53 years old).

A lot of suffering... in our life comes from the desire to change something we can't, so it is just better to let go of that desire (prompted, female, 26 years old).

GROUP OF BELIEFS #4: I would like to, BUT... (denial of control)

This group of beliefs is different in nature compared to previously discussed beliefs, because the assumption underlying these beliefs is that one's personal tourism activity clearly does have negative environmental consequences. To re-establish consonance in this situation requires a different mechanism; in this case, the identification of justifications why, overall, it is unavoidable to engage in these behaviours. The first subset of beliefs is labelled **I have no option**. It assists with reconciliation by arguing that the behaviour is not under one's control for a wide range of possible reasons, for example:

Sometimes I have to go, I have to fly, because it's my business, it's my job and people expect me to be there (prompted, female, 33 years old).

The fact that we live in Australia makes things difficult, I mean, if you want to travel anywhere you need to fly (unprompted, female, 27 years old).

Another way of justifying one's inability to avoid potentially environmentally harmful behaviour is to argue that truly environmentally friendly tourism is very expensive, and **I am not rich enough**. All study participants mentioned budget constraints, for example:

Unfortunately, due to the financial crisis, the budget is becoming more and more of an important constraint, therefore I "allow myself" a lot more than I did years ago (prompted, female, 33 years old).

The following quotes illustrate the same pattern of justifying behaviour:

It's hard... being a budget tourist to actually make mindful decisions (unprompted, female, 27 years old).

The price bracket is only suitable for those with a lot of money. I would say it's really quite elitist! (unprompted, female, 53 years old).

Another belief serving to justify vacation behaviour that is potentially harmful to the environment is that people do not have enough time to do all the research that would be required in order to determine which vacation option has the smallest environmental footprint: **I do not have enough time**.

I had no time to search for a more environmentally friendly option... (unprompted, male, 51 years old).

If... you want to take the train, that will take more time, and you have to have more time (unprompted, female, 33 years old).

Not having enough information or required infrastructure also falls into the group of beliefs relating to denial of control:

I would say in most places it's difficult, because if, for example, water is not filtered so you need to buy bottled water or you want to recycle or compost, but there are no options, or dry clothes but there is only an air dryer (prompted, female, 36 years old).

Hotels don't have recycling facilities (prompted, female, 29 years old).

The reasons why there are no options can be different, ranging from not being in control of the actual travel decision to missing required infrastructure or information about which vacation choices come at the lowest environmental cost. Required infrastructure not being available is a barrier that in many instances actually exists (Juvan & Dolnicar, 2013) and therefore represents both a real barrier (that needs to be removed) and a possible excuse (that needs to be counter-argued with communication messages).

Beliefs in this group represent the construct of perceived behavioural control, as postulated by the theory of planned behaviour.

GROUP OF BELIEFS #5: Vacations are an exception (exception handling)

From early childhood we learn that sometimes it is acceptable to make an exception, to treat oneself to something special. Preschools in Australia teach children that foods fall into two categories: *always foods* and *sometimes foods*, and that we would usually eat *always foods*, but on a special occasion it is acceptable to have *sometimes foods*. Similarly, this group of beliefs is all about arguing that vacation behaviour is *sometimes behaviour*, which is an exception, and therefore the usual behavioural rules do not apply. Exception handling is a term that refers specifically to the management of situations where the standard process is interrupted. In the field of programming, a subroutine is executed that deals with exceptions (for example, Hagen & Alonso, 2000; Strong & Miller, 1995). Once the exceptional situation is over, the standard routine continues. This group of beliefs has been named "exception handling" because the basic operation is similar: people in standard mode, at home, behave in an environmentally friendly way, but on vacation a different "program" is executed.

In this group of beliefs there appear to be two arguments; one is that **vacations are a special treat**, an exception that justifies different behavioural rules:

It's my once-in-a-life chance and I want to do it regardless (prompted, female, 52 years old).

I took five kids to the snow and I know that making snow is not considered as friendly to the environment, but you know what, they enjoyed it (prompted, male, 65 years old).

The second belief in this group is similar, but goes one step further by identifying not only the exceptional status of vacations, but also emphasizing that **usually I am a good person**, thus providing additional justification that it is acceptable to reduce the level of self-discipline in terms of behaviours that have potentially negative environmental consequences. Examples include:

It's like a diet with the 80:20 rule – where for 80% of my diet would be healthy and 20% would be a dessert – it's like that! (prompted, female, 26 years old).

I convince myself I will compensate somewhere else and that no one is perfect (unprompted, male, 26 years old).

I do my part at home where I know what is going on (prompted, female, 31 years old).

GROUP OF BELIEFS #6: Actually, I'm doing more good than bad (compensation through benefits)

The final group of beliefs also acknowledges the negative impact of tourism—and even one's own vacation-related behaviour on the environment—but finds alignment with other attitudes, attitudes which are—maybe temporarily—upgraded in importance to assist in the re-establishment of cognitive consonance. One such attitude is that it is important to assist local communities at the destination:

You can have a negative impact, but if you can help people to potentially become advocates of protection, and these places, you make something good (unprompted, male, 51 years old).

Tourism also does good things, for example, creates jobs, supports conservation, teaches about how to be environmentally friendly, so I guess we can try to compensate that when at home (prompted, female, 36 years old).

All the beliefs discussed above indicate that the modification or addition of beliefs can be used as a mechanism to cope with cognitive dissonance, as postulated by proponents of cognitive dissonance theory. Hardly any evidence could be found in the statements by study participants that the alternative mechanism, that of changing behaviour to cope with dissonance, occurs in the context of vacation-related behaviours. In fact, the only behaviour mentioned by study participants that is clearly driven by concern about the environment and the wish to conserve it (or at least do as little harm as possible) is the voluntary contribution to carbon offset schemes, as illustrated in the following quotes:

I always tick the "offset option" for flying. I know that this probably a bit pathetic, but it makes me feel good (prompted, female, 52 years old).

It is so cheap that I don't trust it, but I am doing it anyway, it seems right for me (unprompted, male, 34 years old).

These findings are in line with the insights reported by Mair (2011): when respondents in her study were asked why they chose to offset, most responses related to either "doing the right thing" or the low cost of purchasing offsets. Although the purchase of carbon offsets is a behaviour, it may in fact act more as an instrument of reconciliation, as suggested by Mair: "some people actively chose to offset in order to act in accordance with their environmental beliefs; other respondents mentioned feeling less guilty about flying" (p. 227).

Conclusions

Tourists generally care about the environment and do not wish to harm it. Yet the very fact that they go on vacation often has negative environmental consequences. The aim of this study is to gain insight into this attitude–behaviour gap. Specifically, we investigated whether people who were active in the protection or conservation of the environment were aware of the negative environmental consequences of tourism, whether there was an attitude–behaviour gap among environmental activists in the tourism context and, if so, how they felt about it and whether they took any action to reduce the gap.

Results provide no evidence that tourists who engaged in environmentalism when at home were unaware of the consequences of tourism in general on the environment, although it became evident in the interviews that they were not sure about the impacts of specific activities relating to vacations. Study participants generally admitted to feeling a tension between their attitudes towards the environment and its protection and their vacation behaviour. This provides evidence of cognitive dissonance occurring in the context of the environmental sustainability of tourism. The conversations with the study participants brought to light a wide range of beliefs that were used to cope with cognitive dissonance, including the denial of the consequences of vacation activities either at the individual level or the level of the tourism industry (as postulated by value-belief-norm theory). Participants also evidenced: downward comparison, which makes their behaviour more acceptable in contrast to worse behaviour by themselves or others (as postulated by social comparison theory); denial of responsibility, either in principle or due to one's powerlessness to make a difference and thus the inability to take responsibility (as postulated by both attribution theory and value-belief-norm theory); denial of control due to external pressures, and financial or time limitations (as postulated by the theory of planned behaviour); exception handling of vacations in contrast to everyday life; and compensation of harm done to the environment through other benefits resulting from tourism.

The identification of these beliefs contributes to our understanding of the attitude–behaviour gap in sustainable tourism, and offers the opportunity for public policy makers and tourism destinations to develop interventions that will target those specific beliefs. Generally, the targeting of beliefs is likely to be more successful than targeting values, which are established early in life and remain largely unchanged. Beliefs are developed and modified throughout a person's life, and therefore represent a more realistic target for causing behavioural change. Possible interventions could, for example, deliberately aim at making it impossible to continue holding certain beliefs, thus actively preventing coping strategies to reduce cognitive dissonance by modifying or adding beliefs. Optimally, the only option for tourists would then be to change behaviour.

Several studies have successfully used people's cognitive dissonance as the target for interventions aimed at behavioural change (Aitken et al., 1994; Dickerson et al., 1992; Kantola et al., 1984). Similar approaches could be developed in the context of environmentally sustainable tourism. For example, people could be provided with information about the extent of the negative impact of a range of typical vacation activities, shown alternatives and asked to make a self-promise to keep the negative environmental cost of their vacation as low as possible. Such

interventions, based on Dickerson et al.'s (1992) experiences, will be most effective when targeted at people who have an awareness of the negative environmental impacts and are self-motivated to keep negative impacts low. Another option would be the elicitation of people's environmental attitudes when checking into a hotel – this could be done by the receptions pointing out the environmental certification of the hotel or by information left for guests to study in the hotel room – and then the use of immediate feedback relating to behaviours with negative environmental consequences, for example, through water and energy meters in hotel rooms. The common core of all such interventions would have to be the deliberate stimulation of cognitive dissonance followed by feedback information on their behaviour. Future work should focus on developing a range of such interventions that specifically target different beliefs identified in this research and used by people to re-establish cognitive consonance, and testing them empirically with actual behaviour, rather than behavioural intentions, as the dependent variable. It will be critical to also determine whether such interventions have negative effects on guest satisfaction or the evaluation of the hotel, which may be negative side-effects of interventions. It should also be noted, that people may find other ways of avoiding to change behaviours, such as shifting from one excuse to another.

The present study is limited in that it focuses on environmental sustainability only. Several study participants raised other aspects of sustainability which they also perceived not to align with their attitudes, such as negative social and cultural impacts on the local communities. It would be interesting to conduct a study focusing on other aspects of sustainability to assess whether tourists use similar mechanisms for re-establishing cognitive consonance.

Appendix

Table 2
Sample sizes in qualitative studies published in annals of tourism research in 2012 and 2013

	Context	Data saturation used	Number of interviews
Andriotis (2013)	Vacation behaviour		26
Berdychevsky and Gibson (2013)	Women's sexual behaviour in tourism	x	21
Cabiddu, Lui, and Piccoli, 2013	Information technology in tourism	x	10
Carnicelli-Filho (2013)	Emotional life of adventure guides		22
Cuffy, Tribe, and Airey (2012)	Education in tourism		65
Daugstad and Kirchengast (2013)	Agri tourism		10
Gao, Zhang, and L'Espoir Decosta (2012)	Perception of destination	x	26
Griffith (2013)	Pilgrimage travel		26
Kidron (2013)	Holocaust tourism		55
Kimbu and Ngoasong (2013)	Tourism development		25
Lai (2013)	Destination representations		28
Lozanski (2013)	Travellers–locals interaction		29
McKercher, Denizci-Guillet, and Ng (2012)	Loyalty in tourism		20
Paraskevas, Altinay, McLean, and Cooper (2013)	Crisis knowledge		21
Pocock and McIntosh (2013)	Notions of home after long trip		5
Shani and Uriely (2012)	VFR tourism		51
Tan, Kung and Luh (2013)	Creative tourism	x	32
Wong et al. (2013)	Buddhism and tourism		24
Average sample size			**28**
Minimum sample size			**5**
Maximum sample size			**65**

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